

Assignment 04

Registration No. : 2025/CS/079

Index No. : 25000799

(01)

```
[chamal@ubuntu:~$ mkdir ~/lab_secure && touch ~/lab_secure/project.txt && chmod 444 ~/lab_secure/project.txt  
[chamal@ubuntu:~$ ls -l lab_secure/  
total 0  
-r--r--r-- 1 chamal chamal 0 Jun 26 15:34 project.txt  
chamal@ubuntu:~$ ]
```

(02)

```
[chamal@ubuntu:~$ cd lab_secure/  
[chamal@ubuntu:~/lab_secure$ touch report.txt log.txt data.csv config.ini  
chamal@ubuntu:~/lab_secure$ ]
```

(03)

```
[chamal@ubuntu:~/lab_secure$ echo -e "Linux File Handling\nPermissions are important\nAccess Control is Key" > report.txt  
[chamal@ubuntu:~/lab_secure$ cat report.txt  
Linux File Handling  
Permissions are important  
Access Control is Key  
chamal@ubuntu:~/lab_secure$ ]
```

(04)

```
[chamal@ubuntu:~/lab_secure$ chmod 640 *.txt  
[chamal@ubuntu:~/lab_secure$ ls -l  
total 4  
-rw-rw-r-- 1 chamal chamal 0 Jun 26 15:35 config.ini  
-rw-rw-r-- 1 chamal chamal 0 Jun 26 15:35 data.csv  
-rw-r----- 1 chamal chamal 0 Jun 26 15:35 log.txt  
-rw-r----- 1 chamal chamal 0 Jun 26 15:34 project.txt  
-rw-r----- 1 chamal chamal 68 Jun 26 15:36 report.txt  
chamal@ubuntu:~/lab_secure$ ]
```

(05)

```
[chamal@ubuntu:~/lab_secure$ ls -l > permissions_check.txt  
[chamal@ubuntu:~/lab_secure$ cat permissions_check.txt  
total 4  
-rw-rw-r-- 1 chamal chamal 0 Jun 26 15:35 config.ini  
-rw-rw-r-- 1 chamal chamal 0 Jun 26 15:35 data.csv  
-rw-r----- 1 chamal chamal 0 Jun 26 15:35 log.txt  
-rw-rw-r-- 1 chamal chamal 0 Jun 26 15:38 permissions_check.txt  
-rw-r----- 1 chamal chamal 0 Jun 26 15:34 project.txt  
-rw-r----- 1 chamal chamal 68 Jun 26 15:36 report.txt  
chamal@ubuntu:~/lab_secure$ ]
```

(06)

```
chamal@ubuntu:~/lab_secure$ mv config.ini ../conf.bak  
chamal@ubuntu:~/lab_secure$ ls .. | grep conf  
conf.bak  
chamal@ubuntu:~/lab_secure$
```

(07)

```
chamal@ubuntu:~/lab_secure$ touch file1.log file2.log && mkdir backup && cp *.log backup/  
chamal@ubuntu:~/lab_secure$ ls backup/  
file1.log file2.log  
chamal@ubuntu:~/lab_secure$
```

(08)

```
chamal@ubuntu:~/lab_secure$ touch report_old.txt && mkdir archives && mv report_old.txt  
archives/report_final.txt  
chamal@ubuntu:~/lab_secure$ ls archives/  
report_final.txt  
chamal@ubuntu:~/lab_secure$
```

(09)

```
chamal@ubuntu:~/lab_secure/testdir$ touch temp1.csv temp2.csv temp_final.csv && ls temp  
*.csv  
temp1.csv temp2.csv temp_final.csv  
chamal@ubuntu:~/lab_secure/testdir$
```

(10)

```
chamal@ubuntu:~/lab_secure$ echo "Hello from file1.txt" > file1.txt && echo "Hello from  
file2.txt" > file2.txt  
chamal@ubuntu:~/lab_secure$ cat file1.txt  
Hello from file1.txt  
chamal@ubuntu:~/lab_secure$ cat file2.txt  
Hello from file2.txt  
chamal@ubuntu:~/lab_secure$
```

(11)

```
chamal@ubuntu:~/lab_secure$ cat file1.txt  
Hello from file1.txt  
chamal@ubuntu:~/lab_secure$ cat file2.txt  
Hello from file2.txt  
chamal@ubuntu:~/lab_secure$ cat file1.txt file2.txt > combined.txt  
chamal@ubuntu:~/lab_secure$ cat combined.txt  
Hello from file1.txt  
Hello from file2.txt  
chamal@ubuntu:~/lab_secure$
```

(12)

(a) Owner has the permission to read and write while Groups and Others only has the read permission

(b) Owner has the permission to read, write and execute. Groups has the permission to read and execute while Others has permission to read only.

(13) Owner will get the permission to read, write and execute. Groups and Others will get the permission to read and execute.

(14)

```
[chamal@ubuntu:~/lab_secure$ mkdir testdir && touch testdir/file1.tmp testdir/file2.tmp
testdir/file3.tmp && mkdir testdir/subdir && touch testdir/subdir/file4.tmp
[chamal@ubuntu:~/lab_secure$ tree testdir/
testdir/
├── file1.tmp
├── file2.tmp
├── file3.tmp
└── subdir
    └── file4.tmp

2 directories, 4 files
[chamal@ubuntu:~/lab_secure$ 
[chamal@ubuntu:~/lab_secure$ rm testdir/*.tmp testdir/subdir/*.tmp
[chamal@ubuntu:~/lab_secure$ tree testdir/
testdir/
└── subdir

2 directories, 0 files
[chamal@ubuntu:~/lab_secure$
```

(15)

```
[chamal@ubuntu:~/lab_secure$ touch script.sh && chmod 700 script.sh
[chamal@ubuntu:~/lab_secure$ ls -l | grep script
-rwx----- 1 chamal chamal    0 Jun 26 16:03 script.sh
[chamal@ubuntu:~/lab_secure$
```

(16)

```
[chamal@ubuntu:~/lab_secure$ echo "Hello Linux" > data.txt && cat data.txt
Hello Linux
[chamal@ubuntu:~/lab_secure$
```

(17)

Absolute path is the directory path from the root directory. Where relative path is the directory path from the current directory.

```
[chamal@ubuntu:~/lab_secure$ cd testdir # relative path
[chamal@ubuntu:~/lab_secure/testdir$ cd ..
[chamal@ubuntu:~/lab_secure$ cd /home/chamal/lab_secure/testdir/ # absolute path
[chamal@ubuntu:~/lab_secure/testdir$
```

(18)

```
chamal@ubuntu:~/lab_secure/testdir$ touch a.sh mann.sh b.txt c.csv && ls ????.sh  
mann.sh  
chamal@ubuntu:~/lab_secure/testdir$
```